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MIL-C-21640A(SHIPS)
12 February 1970
SUPERSEDING
MIL-C-21640(SHIPS)
8 December 1958
(See 6.4)

MILITARY SPECIFICATION

CONTROL, DEGAUSSING COMPASS COMPENSATING, TYPE A-1

1. SCOPE

1.1 This specification covers type A-1 degaussing compass compensating control assemblies which are used to provide control currents to degaussing compass compensating coil assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet, and Strip).

MILITARY

MIL-S-901 - Shock Tests, HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-P-15024 - Plates, Identification --- Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-R-15109 - Resistors and Rheostats, (Naval Shipboard Use).
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-B For Metals).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-J-24142 - Junction Boxes for Electrical Fittings and Fixtures, General Specification For.
MIL-J-24142/3 - Junction Box, Submersible, Size 6 by 9.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.

DRAWINGS

MILITARY

S8106-3232364 - Type A-1 Degaussing Compass Compensating Control Box.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample shall be tested as specified in 4.2 (see 6.2).

3.2 Materials.

3.2.1 The materials shall be as specified herein and shall in all cases be the lightest practicable, consistent with the strength required for sturdiness, serviceability, safety, and reliability.

3.2.2 Nonmagnetic materials. All materials used in this equipment shall be nonmagnetic, (that is, have a permeability less than 1.08 before fabrication and 2.0 after fabrication).

3.3 Design. The design, layout, and assembly of the units and their component parts shall be such as to facilitate quantity production. Jigs shall be used to position parts during welding and assembly operations.

3.4 Component parts. The type A-1 degaussing compass compensating control assembly shall be in accordance with Drawing S8106-3232364 and shall consist of six resistor assemblies (see 3.9) and one 8 terminal, terminal board, located within a lightweight, watertight enclosure (see 3.10).

3.4.1 Size and weight. In the design of all components emphasis shall be placed upon reduction of the weight of the unit. Mounting and overall dimensions shall comply with Drawing S8106-3232364.

3.5 Painting. After all machining, welding, and brazing operations are completed, the exterior and interior surfaces of all enclosures shall have all rust or other visible corrosion products removed; shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing; and shall then be painted as specified in 3.5.1 and 3.5.2.

3.5.1 Primer. One coat of primer, manufactured in accordance with MIL-P-15328 shall be applied as a continuous film 0.0002 to 0.0005 inch thick.

3.5.2 Enamel. Two coats of gray enamel, conforming to type II, class 2 of MIL-E-15090, shall be applied as continuous films approximately 0.001 inch thick. Class 2 (semigloss) shall be applied on permanently installed equipment and class 1 (gloss) followed by a heavy coat of hard wax shall be applied on portable equipment.

3.6 Rounded corners and edges. All edges and corners of external surfaces, normally exposed or to be painted, shall be rounded. Sharp surfaces of any kind shall be avoided.

3.7 Insulation distances. The minimum creepage and clearance distances between electric circuits or between any electric circuit and ground shall be as follows:

Clearance in air - 1/4 inch
Surface creepage - 1/4 inch

Creepage and clearance distances are defined in MIL-E-917.

3.8 Wiring. Internal wiring shall be protected with insulation capable of withstanding continuous heat dissipation, from each of the three circuit assemblies, of 150 watts with all three circuits operating simultaneously. All wiring shall be protected and secured to prevent chafing of insulation of any of the conductors due to vibration. Electrical insulation shall be capable of satisfactory service with the heat dissipation specified herein.

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3.9 Resistor assemblies. The control assembly shall consist of six resistor assemblies comprised of three "coarse adjustment" variable resistors and three "fine adjustment" variable resistors. The resistors shall be grouped to form three circuits. Each circuit shall include one "coarse adjustment" and one "fine adjustment" resistor. Circuitry and circuit components, and assemblies shall be designed in accordance with and interchangeable with those shown on Drawing S8106-3232364.

3.9.1 Resistors. The resistors used in each resistor assembly shall conform to MIL-R-15109. They shall be designed for continuous duty and shall be capable of operating under rated conditions.

3.9.1.1 The electrical and mechanical characteristics of the resistor units shall approximate that shown in table I.

Table I - Electrical and mechanical characteristics.

Resistor	Tube dimensions (inches)			Allowable resistance variations	Ohms per winding	Number turns per	Length of tube covered by winding	Winding ampere capacity (continuous)
	Length	O.D.	I.D.					
				Percent			Percent	
15 ohm	2-3/4	9/16	3/8	+10	15	55	100	2.0
300 ohm	2-3/4	1-1/8	3/4	-10 +20	260	77	50	0.40
					40	33	50	1.30

3.9.1.2 Terminal block. An 8 terminal, terminal block shall be provided in the control assembly. Materials used in this block shall be capable of withstanding the heat dissipation outlined in 3.8. Creepage and clearance distances involved shall be in accordance with 3.7. Each terminal shall be marked to indicate its function in the electrical circuit. The terminal board shall have an identification strip and shall be marked in accordance with Drawing S8106-3232364.

3.10 Enclosure.

3.10.1 Size, weight and material. The enclosure shall be constructed in accordance with MIL-J-24142 and MIL-J-24142/3. The enclosure material shall be brass in accordance with QQ-B-613. Emphasis shall be placed on making this enclosure as lightweight as practicable.

3.10.2 Degree of enclosure. The degree of enclosure shall be "watertight" as specified in MIL-STD-108.

3.10.3 Enclosure mounting. The enclosure shall be designed for bulkhead mounting. Mounting dimensions shall be in accordance with Drawing S8106-3232364.

3.10.4 Accessibility. The enclosure shall be designed for maximum accessibility for repair and replacement of parts. The resistor assemblies shall be so mounted as to permit replacement of those components likely to become defective without extensive disassembly of other parts and assemblies within the enclosure.

3.10.5 Cover bolts. The enclosure cover bolts shall be nonremovable.

3.10.6 Condensation drain. A removable plug shall be provided in the bottom of the enclosure for draining any condensate.

3.10.7 Terminal tubes. Terminal tubes will not be provided by the manufacturer.

3.10.8 Identification plate. A type A identification plate in accordance with MIL-P-15024 shall be mounted on the outside of the enclosure cover. The identification plate shall be in general accordance with the sample shown on figure 2A of MIL-P-15024 and shall contain the following information:

(a) Compass compensating control box, type A-1.
(b) Federal stock number H1075-511-7174.
(c) Manufacturer.
(d) Contract number.

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3.11 Temperature and humidity. The type A-1 control assembly shall be capable of satisfactory, continuous operation in any ambient temperature between the limits of -20 degrees Fahrenheit(°F.) and 180°F. and in any relative humidity of 95 percent or less. This equipment shall be suitable for extended storage in any ambient temperature between the limits of -65°F. and 180°F.

3.12 Salt spray. The equipment shall be capable of withstanding the salt spray test specified in 4.5.4.

3.13 Dielectric strength. The equipment shall be capable of withstanding the dielectric test voltages specified in 4.5.5.

3.14 Vibration. The equipment shall be capable of withstanding the vibration test specified in 4.5.6 without mechanical or electrical damage.

3.15 Shock. The equipment shall be capable of withstanding the HI shock test specified in 4.5.7 without mechanical or electrical damage.

3.16 Designation and marking. A schematic wiring diagram and an operational information plate in accordance with MIL-P-15024 shall be permanently attached to the exterior of the enclosure cover, silk screening or metal photo process is permitted. These items shall be protected against attack from salt spray. All data and information required to be placed on components by Drawing S8106-3232364 may be applied by silk screening. Each resistor assembly shall be marked "fine adjustment" or "coarse adjustment" as applicable.

3.17 Repair parts. Repair parts shall consist of one each of parts 24 and 25 shown on Drawing S8106-3232364 and four each of parts 26 and 27 shall be furnished for each complete assembly.

3.18 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454. The equipment shall comply with the visual examination specified in 4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in 4.4 and 4.5.1 through 4.5.8.

4.3 Quality conformance inspection. Each unit offered for delivery on the contract or order shall be subjected to the examination and tests specified in 4.4, 4.5.1 and 4.5.5.

4.4 Visual examination. The completed equipment shall be given a thorough visual examination to determine that it conforms to the applicable specifications and approved working drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight and marking of identification and information plates. The examination shall be limited to those tests that may be performed without disassembling the component in such a manner that its performance, durability or appearance would be affected. The visual examination shall include a check of all operating controls, circuit functions, adjustments, and an operating test of the complete equipment (see 4.5.8).

4.5 Test methods.

4.5.1 Creepage and clearance distance tests. Creepage and clearance distances shall be demonstrated by actual measurement and shall conform to the requirements of 3.6.

4.5.2 Enclosure tests. The effectiveness of the component enclosures shall be tested for watertightness (3 foot). The test for waterproof enclosures shall be as specified in 4.5.2.1 through 4.5.2.4.

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4.5.2.1 The enclosure shall be submerged under a 3 foot head of standard sea water specific gravity (sp. gr.) 1.025 at approximately 15 degrees Centigrade(°C.) for a period of 1 hour.

4.5.2.2 After immersion for 1 hour the equipment shall be removed from the water and shall be permitted to drain for a period of one hour while being held in the position in which it would normally be mounted onboard ship.

4.5.2.3 The equipment shall then be energized in the normal manner for a period of one hour.

4.5.2.4 Four hours after the equipment has been removed from the water it shall be disassembled and inspected. There shall be no evidence of water pockets and no evidence of corrosion.

4.5.3 Temperature and humidity tests. The equipment shall be operated in the ambient temperatures and humidity as specified in 3.11 to determine any deleterious effect therefrom. The test unit shall be subjected to a series of 15 temperature cycles within the range of temperatures specified in 3.11. Temperature tests shall be conducted in accordance with methods 501 and 502 of MIL-STD-810. Humidity test shall be conducted in accordance with method 507 of MIL-STD-810.

4.5.4 Salt spray test. A salt spray test shall be conducted in accordance with method 101 of MIL-STD-202, test-condition letter B.

4.5.5 Dielectric strength test. A dielectric strength test shall be conducted in accordance with method 301 of MIL-STD-202. The test shall be performed by applying a test voltage of 1500 volts (rms) between the electrical circuits and the enclosure.

4.5.6 Vibration. Vibration tests shall be performed on the completed assembly in accordance with the type I environmental test of MIL-STD-167. After completion of tests the assembly shall be examined for evidence of leakage or physical damage.

4.5.7 Shock. Shock tests shall be conducted on a shock-testing machine for lightweight equipment in accordance with grade A, class I, type A of MIL-S-901 (see 3.15). Failure which is defined as inability to perform principal functions shall result in rejection. Failure to perform principal functions is defined as follows:

(a) Breakage of any parts, including mounting bolts. Minor chipping or distortion of parts will be accepted when such chipping or distortion cannot in any way impair the operation or performance requirements of the equipment.
(b) Appreciable distortion of any parts including the case and mounting appendages.
(c) Insulation resistance values lower than those permitted by this specification.
(d) Failure to pass the dielectric strength test.
(e) Failure to pass the operating test.

4.5.8 Operating test. The unit shall be given an operating test as follows: A 6-ohm load resistor shall be connected across each of the three pairs of output terminals and the unit then connected to a 12-volt d.c. power source. Each of the three individual circuits in the unit shall then be adjusted to simultaneously supply a 1.0 ampere output current to the 6-ohm load resistor. The unit shall then be operated in this condition, with the cover closed, for a period of 2 hours. The unit shall be observed during this test for any evidence of hot spots or other indication of unsatisfactory operation.

4.6 Test reports. When specified (see 6.1), a test report, certified by the cognizant Defense Contract Administration Services Office (DCASO), shall be forwarded in triplicate to the command or agency concerned to substantiate the fact that tests have been conducted and the specification requirements have been met.

4.7 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

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5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts (see 5.3.1).

5.1.1 Control assemblies.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged level A, packed level C and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.3.2 and 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation, packaging, packing and marking. The equipment, accessories, technical manuals and repair parts shall be preserved and packaged by level A or C; packed by level A, B or C and marked as specified in the contract or order in accordance with MIL-E-17555. Method II shall apply for level A preservation and packaging of the equipment.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The required delivery dates for parts and equipment.
(c) Type and quantity of repair parts (see 3.17).
(d) Quality assurance requirements (see 4.1.1).
(e) When test reports are required (see 4.6).
(f) Preservation, packaging, packing and marking requirements other than those required by 5.1 (see 5.2).
(g) When polystyrene (loose-fill) material is approved (see 5.3.1).

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6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Navy - SH
(Project 1075-N017)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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